



LLM-augmented multi-agent cooperative framework for medical case retrieval in cardiology

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Abstract

Retrieving relevant medical cases or documents is a critical information retrieval (IR) task in clinical decision support, particularly in cardiology, yet traditional search methods struggle with complex semantic queries in healthcare. Recent advances in large language models offer powerful language understanding, but LLMs alone cannot reliably retrieve factual cases due to knowledge cutoffs and hallucinations. We focus on a case retrieval task – given a textual description of a patient case, find similar prior cases or pertinent literature – formulated as a general IR problem rather than a purely medical study. Conventional lexical methods often miss semantic similarities, while static dense retrievers falter on out-of-domain medical vocabularies. LLMs can comprehend queries and context, but without external knowledge they may produce inaccurate or non-transparent results. We propose a novel LLM-augmented multi-agent retrieval framework that marries an LLM with dedicated retrieval agents in an iterative cooperation mechanism. Our method uses a LLM as a “planner” agent to reformulate queries and integrate medical (e.g., cardiology) context, and a retrieval agent (with a knowledge index) to fetch candidate cases; the agents interact iteratively, refining search and reranking results via a retrieval-augmented generation (RAG) loop. This multi-agent design contributes three innovations: (1) an iterative query refinement strategy guided by LLM reasoning chains; (2) a cooperative retrieval architecture where an LLM agent and a search agent exchange information to improve relevance; (3) an LLM-based relevance estimator that grounds the LLM with retrieved evidence to mitigate hallucinations. Experiments on three open medical text datasets show our method outperforms baseline models by 5.3–6.1 percentage points in Recall@10 and NDCG, with statistically significant gains. We also observe improved generalization to novel conditions and robustness to query noise compared to baselines. The proposed framework, while validated on medical text, is broadly applicable to other knowledge-intensive retrieval tasks (legal case search, technical support archives), providing a foundation for intelligent IR systems that leverage both learning-based understanding and explicit retrieval for transparency and up-to-date knowledge.

Keywords Natural language processing · Multi-agent retrieval · Large language models · Retrieval-augmented generation · Medical information retrieval

1 Introduction

Retrieving relevant textual information is a fundamental problem in computer science, with applications ranging from web search to recommender systems. In specialized

domains like healthcare, medical case retrieval – finding similar patient cases or pertinent clinical literature given a query case – is crucial for evidence-based medicine (Tsatsaronis et al. 2015) and clinical decision support (Roberts et al. 2020). This task can be cast as an information retrieval (IR) problem over unstructured text (electronic health records, case reports, Q&A logs, etc.), and it poses unique challenges to conventional IR techniques. Traditional keyword-based methods (e.g. BM25) often fail to capture the semantic and contextual richness of clinical language (Robertson et al. 2009). For example, a query describing “chest pain and diaphoresis in a 55-year-old” might not explicitly match documents containing “myocardial infarction symptoms,” yet a human expert recognizes them as related – a gap that simple

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lexical matching misses. Neural IR models have improved semantic matching by encoding text into vector representations (Karpukhin et al. 2020; Reimers and Gurevych 2019). BERT-based retrievers and re-rankers can capture contextual similarity beyond exact words (Nogueira and Cho 2019; Khattab and Zaharia 2020). However, these models typically require large training data with relevance labels and struggle when applied to medical text without domain adaptation. Domain-specific language models like BioBERT and ClinicalBERT (Lee et al. 2020; Alsentzer et al. 2019) partially address medical terminology, but they may not generalize well to rare conditions or new medical findings, leading to suboptimal recall in retrieval tasks on medical data (Hambarde and Proenca 2023).

Meanwhile, large language models (LLMs) such as GPT-3 and GPT-4 have demonstrated remarkable capabilities in understanding and generating human-like text (Brown et al. 2020; Bubeck et al. 2023). Notably, LLMs are few-shot learners that can perform complex tasks with minimal task-specific training (Brown et al. 2020). In the medical domain, LLMs have shown the ability to encode a surprising amount of clinical knowledge (Singhal et al. 2023) and answer medical questions at an expert level in certain evaluations (Singhal et al. 2023; Wang et al. 2024). This suggests they grasp domain semantics well. However, using LLMs alone for retrieval is problematic. LLMs are generative and lack direct access to up-to-date external knowledge, relying only on their training data (Ji et al. 2023). They often produce hallucinated content – plausible-sounding but incorrect information (Bubeck et al. 2023) – which is unacceptable when querying for evidence in healthcare. Additionally, LLMs do not inherently return sources or document identifiers for their answers, limiting transparency. Simply asking an LLM to recall similar cases is akin to a “closed-book” approach and will miss relevant cases not seen during training. Therefore, LLMs cannot replace explicit retrieval engines when accurate and verifiable case lookup is required (Ji et al. 2023; Lewis et al. 2020).

A promising direction is to integrate LLMs with retrieval – harnessing the comprehension and reasoning abilities of LLMs together with the grounding of IR systems. Retrieval-Augmented Generation (RAG) techniques have emerged to incorporate external documents into LLM outputs (Lewis et al. 2020; Gao et al. 2023). By fetching relevant text snippets and feeding them into the LLM’s context, RAG can improve factual accuracy and reduce hallucinations in domains like healthcare (Gao et al. 2023; Zhao et al. 2024). For instance, Lewis et al. (2020) demonstrate that querying a knowledge base for supporting evidence significantly boosts performance on knowledge-intensive tasks. However, most existing RAG frameworks treat retrieval as a single-

step tool call: the LLM issues one query to a search module, gets some documents, and then generates an answer (Lewis et al. 2020; Yao et al. 2023). This one-shot approach may be insufficient for complex case queries that require iterative exploration. In medical case retrieval, a query might be initially broad or underspecified – e.g., “atypical chest pain case” – requiring follow-up clarification (Which features are most relevant? What specific context?). Humans often conduct multi-turn searches, reformulating queries after seeing initial results. Recent work on LLM-based agents suggests that LLMs can emulate such interactive query reformulation. Yao et al. (2023) introduced ReAct, where the LLM interleaves reasoning and tool use, enabling multi-step query-answering. Similarly, Press et al. (He et al. 2025) and others propose letting the model ask itself follow-up questions (self-asking) and refine queries iteratively. These approaches hint that an LLM could guide a search process through multiple turns, acting as a high-level decision-maker or “agent” that decides when and how to query the data.

In parallel, the concept of multi-agent systems with LLMs is gaining traction (Shen et al. 2023; Nachimovsky et al. 2025). Instead of a single monolithic model, multiple specialized agents (which could be instances of LLMs or a mix of models) can collaborate on sub-tasks, communicating in natural language. For example, one agent might specialize in planning and another in execution (Shen et al. 2023). In IR, a recent perspective by Nachimovsky et al. (2025) advocates modeling retrieval as interactions between query agents, document agents, and ranker agents. This is particularly relevant in dynamic environments where both queries and documents might be generated or analyzed by AI. A multi-agent viewpoint allows us to break the complex retrieval problem into modular components: one agent formulates refined queries, another fetches documents, another evaluates them, etc., all coordinated towards a common goal. Such a framework can potentially capture reasoning loops and negotiation that a single-step method cannot (Nachimovsky et al. 2025). While multi-agent IR is still nascent, early explorations show improved performance when agents handle different aspects of the search process (Nachimovsky et al. 2025).

Considering these developments, we identify a critical research gap: existing IR methods (both classical and neural) struggle with the semantic and generalization demands of medical case retrieval, and naive LLM integration (single-step RAG) does not fully leverage the LLM’s reasoning potential. There is a need for an innovative computing method that brings together the strengths of LLMs and IR in a more synergistic, interactive manner. Our work addresses this by introducing a cooperative multi-agent retrieval framework centered on LLM guidance, with the medical case domain as

a challenging application to validate its effectiveness. Importantly, our focus is on methodological innovation in IR and NLP (the medical aspect is a testbed, not the primary contribution).

We explicitly highlight the key methodological contributions of our study:

1. We propose a novel retrieval framework where a planner agent (powered by an LLM) and a search agent (IR module) collaborate in multiple rounds to retrieve relevant cases. Unlike one-shot RAG, our architecture enables iterative querying and dynamic feedback.
2. We develop a technique for iterative query refinement using LLM reasoning. The LLM agent generates clarifying questions and reformulated queries based on initial results, akin to a human search strategist. We incorporate a chain-of-thought prompting scheme in the LLM to allow it to explain the query logic and determine what to search next. This leads to more effective searches, especially for complex medical queries with multiple facets.
3. At the core of our retrieval agent, we employ a hybrid model that combines dense semantic retrieval with lexical matching. We fine-tune a bi-encoder model on medical text for encoding cases, ensuring domain-specific terminology is understood. Additionally, we integrate the LLM agent as a reranker that evaluates retrieved candidates in natural language. By fusing model-driven relevance scoring with LLM-based judgment, our method achieves high recall and precision, addressing the domain adaptation issue in prior work.
4. We conduct comprehensive experiments on three benchmark datasets covering clinical narratives and biomedical literature. We compare against a broad spectrum of 10+ baselines, results show consistent improvements of 4–8% absolute on ranking metrics (NDCG, Recall), demonstrating the efficacy of our approach. We also perform ablation studies to quantify the contribution of each component, cross-domain generalization tests, and robustness checks. Furthermore, we analyze case studies to illustrate how the LLM agent's explanations enhance interpretability, a crucial aspect for clinical adoption.

Overall, our work advances the state-of-the-art in information retrieval by introducing a cooperative LLM-agent paradigm that is particularly beneficial for knowledge-intensive domains. While validated on medical case retrieval, the proposed techniques are general and lay the groundwork for applying multi-agent LLM-augmented retrieval in other domains (legal case retrieval, technical support archives, etc.), aligning with the broader push towards explainable and generalizable AI systems in IR.

2 Related work

2.1 Neural information retrieval and case-based search

Traditional vs Neural IR Early IR methods relied on lexical matching and probabilistic models. BM25 remains a strong classical baseline, scoring documents by term frequency and inverse document frequency (Robertson et al. 2009). However, BM25 cannot capture semantic similarity beyond exact token overlap. The advent of deep learning brought neural IR models that encode text in vector spaces to measure semantic relevance (Reimers and Gurevych 2019; Mitra et al. 2018). For instance, the Deep Relevance Matching Model (DRMM) introduced by Guo et al. (2016) applied neural networks to model term-level interactions for ad-hoc retrieval. More recent approaches learn end-to-end text representations: DPR (Dense Passage Retrieval) uses twin BERT encoders for questions and passages, trained on a contrastive loss to retrieve Wikipedia passages for QA (Karpukhin et al. 2020). ColBERT refines this with late interaction, computing fine-grained term embeddings and achieving efficient yet accurate passage search (Khattab and Zaharia 2020). Extensive benchmarks like MS MARCO and BEIR have driven progress in neural IR, revealing that dense retrievers often outperform sparse methods on semantic matching tasks (Thakur et al. 2021). In the biomedical domain, neural IR has been applied to literature search and patient record retrieval. For example, BioBERT-based retrievers leverage domain-pretrained BERT to encode medical texts (Lee et al. 2020), improving over generic BERT on biomedical articles. Despite their success, neural retrievers face challenges: they require large training sets of relevant query-document pairs, and performance can degrade when queries are from a different distribution than training data (the domain shift problem). In medical IR, labeled data is scarce and expensive to obtain, as relevance judgments need expert domain knowledge. Some unsupervised or zero-shot techniques have been explored, such as Contriever which uses contrastive learning on unlabeled text to build general-purpose dense retrievers (Gao et al. 2023). Gao et al. (2023) show that with careful pre-training (e.g. inverse cloze tasks and corpus-level MLM), zero-shot dense retrieval improves, but there remains a gap when tackling highly specialized queries.

Case-based medical retrieval Retrieving similar patient cases or relevant medical literature has been an area of interest in biomedical informatics for decades. Early systems used structured data or simple text filters to find matching records. The Text Retrieval Conference held a Clinical Decision Support (CDS) Track (2014–2016) specifically targeting case-based retrieval: given a medical case narrative, the task was to retrieve relevant journal articles for answering clin-

ical questions (diagnosis, test, treatment) (Tsatsaronis et al. 2015). Participants applied variations of BM25, knowledge-driven query expansion, and later, initial neural rerankers as BERT emerged. Simpson et al. (Kirk et al. 2016) and Roberts et al. (Kirk et al. 2016; Roberts et al. 2020) provide overviews of these efforts. Results indicated that lexical methods with domain-specific expansions (e.g., adding MeSH terms) were effective baselines, but complex narratives remained challenging for automated systems. More recently, researchers have begun applying modern neural IR to clinical data. For example, dense embedding models have been used to identify similar patients in electronic health records for cohort study support (Miotto et al. 2016). Miotto et al. (2016) learned unsupervised patient representations (“Deep Patient”) from EHRs, which could be used to measure patient similarity, though their focus was predictive modeling rather than retrieval of records. Other works use deep metric learning to explicitly learn a patient similarity function: for instance, convolutional neural networks over structured and unstructured data to cluster similar patients (Thakur et al. 2021; Devlin et al. 2019) (ref. patient similarity networks). However, these often treat similarity as a supervised classification/regression (predict some outcome or match), as opposed to actual retrieval ranking. Our work differs by directly tackling unstructured textual case retrieval with a focus on generalization and using minimal supervised data. We also emphasize interpretability; whereas a metric learning model might give a similarity score, our LLM agent can provide natural-language rationales (e.g., “both patients have a history of smoking and present with early-onset emphysema”), aligning with the need for explainability in clinical tools (Achiam et al. 2023; Touvron et al. 2023; Zakka et al. 2024).

2.2 Large language models and retrieval augmentation

LLMs in knowledge-intensive tasks Large language models like GPT-3 (Brown et al. 2020) and GPT-4 (Bubeck et al. 2023) have transformed NLP by demonstrating strong performance in question answering, summarization, and dialogue through few-shot prompts. However, for tasks requiring up-to-date or detailed knowledge (e.g., “What is the latest guideline for treating condition X?”), LLMs struggle because their knowledge is frozen at training time and may be incomplete or outdated (Ji et al. 2023). Researchers have thus explored retrieval-augmentation: combining LLMs with IR to ground their responses. Lewis et al. (2020) introduced Retrieval-Augmented Generation (RAG), where an LLM-based generator is paired with a neural retriever to incorporate external documents into answer generation. RAG and related frameworks like REALM and FiD (Fusion-in-

Decoder) have shown significant gains in open-domain QA, fact-checking, and knowledge-intensive dialog by mitigating hallucinations and improving factual correctness (Lewis et al. 2020; Gao et al. 2023; Jeong et al. 2024). In the medical domain, Singhal et al. (2023) presented Med-PaLM (an adaptation of GPT-4) and found that providing retrieval capability could help the model handle questions it otherwise answered incorrectly. A recent survey by Gao et al. (2023) systematically reviews RAG techniques for LLMs, classifying approaches into naïve RAG (single-pass retrieve and read) versus more advanced iterative retrieval and modular generation. For instance, one study proposes iterative RAG for medicine, where an LLM can ask follow-up queries based on initial results (Zhao et al. 2024) – this resonates with our approach of multi-turn retrieval. Another survey by Zhao et al. (2024) discusses RAG for ensuring the provenance of AI-generated content, reinforcing that retrieval helps in generating verifiable and trustworthy outputs. Our work builds on these insights, extending the paradigm by using the LLM not just as a passive consumer of retrieved facts, but as an active controller that drives the retrieval process itself (Pal et al. 2023; Xiong et al. 2024; Schick et al. 2023; Wei et al. 2022).

Interactive & multi-step retrieval with LLMs The idea of LLMs actively querying and interacting with tools has been explored in several contexts. The ReAct framework (Yao et al. 2023) demonstrated that an LLM (prompted suitably) can produce a sequence of reasoning statements and tool-use actions, enabling it to perform multi-step searches or API calls. For example, given a complex question, a ReAct agent might issue a search query, read the results, then reason and search again, emulating human research behavior (Yao et al. 2023). Similarly, Self-Ask prompting lets an LLM break a question into sub-questions and answer them one by one (possibly using a search engine for each) (He et al. 2025). These approaches showed improved performance on multi-hop QA and logical reasoning tasks by decomposing problems. In our scenario, a medical case query might implicitly be multi-hop (symptoms -> possible conditions -> similar case reports); an LLM that can iteratively search should handle this better than one-shot retrieval. Beyond single-agent prompting, recent systems use multiple LLMs or modules in a pipeline or loop. Shen et al. (2023) introduced HuggingGPT, where ChatGPT acts as a coordinator that delegates tasks (e.g., image analysis, speech recognition) to appropriate expert models. While their focus was multi-modal AI tasks, the concept of an LLM as a central planner is analogous to our planner agent. There is also growing interest in multi-agent communication: could two or more LLM agents converse to solve a problem? For instance, one agent could play the role of a “user” or “questioner” asking clarifications, and another the “expert” answering, thereby refining

the query through dialogue. This approach was explored in dialog systems and brainstorming sessions between agents, showing potential to improve quality via iterative feedback. Park et al. and others have even simulated communities of agents to collaborate on tasks (e.g., Generative Agents) – albeit not for retrieval tasks (Nachimovsky et al. 2025). Our framework can be seen as a specific instantiation of this trend: we effectively create an agent that asks implicit questions (via query reformulation) and another that provides answers (via retrieval results), iterating until satisfied (Mialon et al. 2023; Lazaridou et al. 2022; Chen et al. 2017; Qu et al. 2020; Zhao et al. 2023).

Domain adaptation & generalization A notable advantage of integrating retrieval with LLMs is improved domain adaptability. Pure LLM solutions struggle with domain-specific jargon (e.g., medical abbreviations) unless fine-tuned extensively on that domain (Alsentzer et al. 2019; Gu et al. 2021). In contrast, retrieval can bring in actual domain text as needed. Domain-specific models like PubMedBERT (Gu et al. 2021) and ClinicalBERT (Alsentzer et al. 2019) were developed to address this by pretraining on biomedical corpora. They have been used as backbones for many medical NLP tasks and typically outperform generic models on in-domain evaluations. We leverage such models in our retrieval backbone to initialize semantic embeddings. However, even these models may not encode clinical reasoning out of the box (they learn language patterns, not how to connect a patient case to similar cases). There have been efforts to inject structured medical knowledge or train models to cite evidence (Jeong et al. 2024). For example, Jeong et al. (2024) combine retrieval with a self-reflection step for medical QA, showing that retrieving relevant facts and then letting the model reason yields better medical reasoning accuracy. Another work by Nachimovsky et al. (2025) used retrieval to help an LLM explain medical answers in lay language, essentially augmenting domain knowledge. These indicate that hybrid approaches can achieve both accuracy and explainability, which purely neural end-to-end models lack. Our approach further emphasizes explainability: by having an LLM agent articulate why it is refining a query or why a retrieved case is relevant, we produce a trace of reasoning. This is important for user trust, especially in medicine where clinicians must understand the basis of recommendations. Prior IR systems for healthcare rarely provided such reasoning, other than simple keyword highlighting. In contrast, our multi-agent system can generate a summary of how a past case is analogous to the query case, offering a form of case-based explanation (Formal et al. 2021; Santhanam et al. 2021; Nogueira et al. 2019; Dai et al. 2022).

In summary, our work is grounded in several strands of recent research: neural IR for semantic matching, LLM inte-

gration for knowledge augmentation, interactive multi-step retrieval, and domain adaptation in biomedical NLP. By synthesizing these directions, we aim to overcome their respective limitations—such as the rigidity of traditional IR, the isolation of LLMs from external data, and the brittleness of single-step pipelines. While prior work has explored individual aspects like ReAct-style reasoning, self-ask prompting, multi-agent retrieval frameworks, and LLM-based reranking, our method is among the first to integrate these components into a unified, cooperative multi-agent retrieval framework. We further validate this approach systematically on challenging medical case retrieval benchmarks, highlighting its practical benefits in this domain. In the next section, we detail our methodology, followed by an extensive experimental evaluation that substantiates the benefits of our approach over existing methods.

3 Methodology

In this section, we formalize the medical case retrieval task and describe our LLM-augmented multi-agent retrieval framework in detail. Our method consists of three main modules – the Query Planner (LLM Agent), the Retrieval Module, and the Relevance Evaluator – that work together in an iterative loop. We first present the task definition and notation, then explain the overall architecture with algorithmic pseudocode, and finally delve into each component’s technical design, including model training objectives and integration strategies. Key novel aspects, such as the multi-turn query refinement and LLM-based reranking, are highlighted.

3.1 Task definition and notation

We define medical case retrieval as follows: let $D = d_1, d_2, \dots, d_{|D|}$ be a corpus of textual documents representing cases or articles (e.g. patient discharge summaries, case reports, Q&A snippets). Given a query q which is a textual description of a patient’s case or an information need (e.g. “55-year-old male with atypical chest pain and mild troponin elevation”), the goal is to retrieve a ranked list of documents $[d_{(1)}, d_{(2)}, \dots, d_{(k)}]$ from D that are relevant to q . We assume relevance is defined in a task-specific way (for instance, documents about patients with the same diagnosis or containing useful insights for the query case). In an ideal scenario, a ground truth set $R(q) \subseteq D$ of relevant items exists for evaluation. The retrieval system’s job is to produce an ordering of documents that places $R(q)$ at the top. Formally, we aim to learn a scoring function $s(q, d)$ such that for any two documents d_i, d_j , if d_i is more relevant to q than d_j , then $s(q, d_i) > s(q, d_j)$. At query time, the system returns the top- K documents by this score:

$$\text{RetrieveTopK}(q; D) = \arg \max_{D'} \sum_{d \in D'} s(q, d) \quad (1)$$

where D' ranges over all size- K subsets of D . In practice, exhaustive maximization is infeasible for large D , so approximate methods (indexing, search algorithms) are used to compute this top- K set efficiently.

A classical approach is to design $s(q, d)$ as a similarity measure (cosine similarity, dot product, etc.) between some representation of q and d . Traditional IR uses sparse vectors (based on term frequencies), while modern neural IR uses dense vectors from learned encoders. Let $f_\theta(q) \in \mathbb{R}^m$ and $g_\theta(d) \in \mathbb{R}^m$ denote the query and document embeddings produced by a neural bi-encoder model with parameters θ . A common choice is:

$$s_{\text{dense}}(q, d) = \langle f_\theta(q), g_\theta(d) \rangle, \quad (2)$$

the dot product of the embeddings (or cosine similarity, which is a scaled version if vectors are L_2 -normalized). The system can then retrieve the top results by nearest-neighbor search in the embedding space. We additionally consider the BM25 score as a complementary signal:

$$s_{\text{BM25}}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1(1 - b + b \frac{|d|}{\text{avgdl}})}, \quad (3)$$

where $f(t, d)$ is the term frequency of term t in d , and k_1 , b are hyperparameters. This is the standard BM25 formula capturing term matching with length normalization.

In our hybrid approach, we combine dense and sparse scores:

$$s_{\text{hybrid}}(q, d) = \lambda \cdot \frac{s_{\text{dense}}(q, d)}{\|f_\theta(q)\| \|g_\theta(d)\|} + (1 - \lambda) \cdot \frac{s_{\text{BM25}}(q, d)}{\max_{d' \in D} s_{\text{BM25}}(q, d')}, \quad (4)$$

where we normalize each score and $\lambda \in [0, 1]$ controls the weight (we set $\lambda \approx 0.5$ based on validation). The hybrid score exploits both semantic matching from the neural model and exact term evidence from BM25.

The task is complicated by the fact that q may not be a perfect description of what to search for. Users (or patients/clinicians) might omit details or use non-standard terms. Thus, an implicit part of the task is query understanding and reformulation. We treat the initial query q as potentially requiring transformation into an improved query q' that yields better results. For example, $q = \text{"chest pain with normal angiogram"}$ might be reformulated to $q' = \text{"MINOCA (myocardial infarction with non-obstructive coronary arteries) case"}$. We denote a sequence of query states $q^{(0)}, q^{(1)}, \dots, q^{(T)}$ where $q^{(0)}$ is the original query and $q^{(T)}$ is the final refined query after T iterations (or T can be dynamically chosen when the system decides to stop). The relationships between these queries and retrieved sets define the multi-turn process.

3.2 Overall framework architecture

Figure 1 illustrates the architecture of our multi-agent retrieval framework. The system comprises two primary agents: (A) the LLM Planner Agent and (B) the Search/Re-

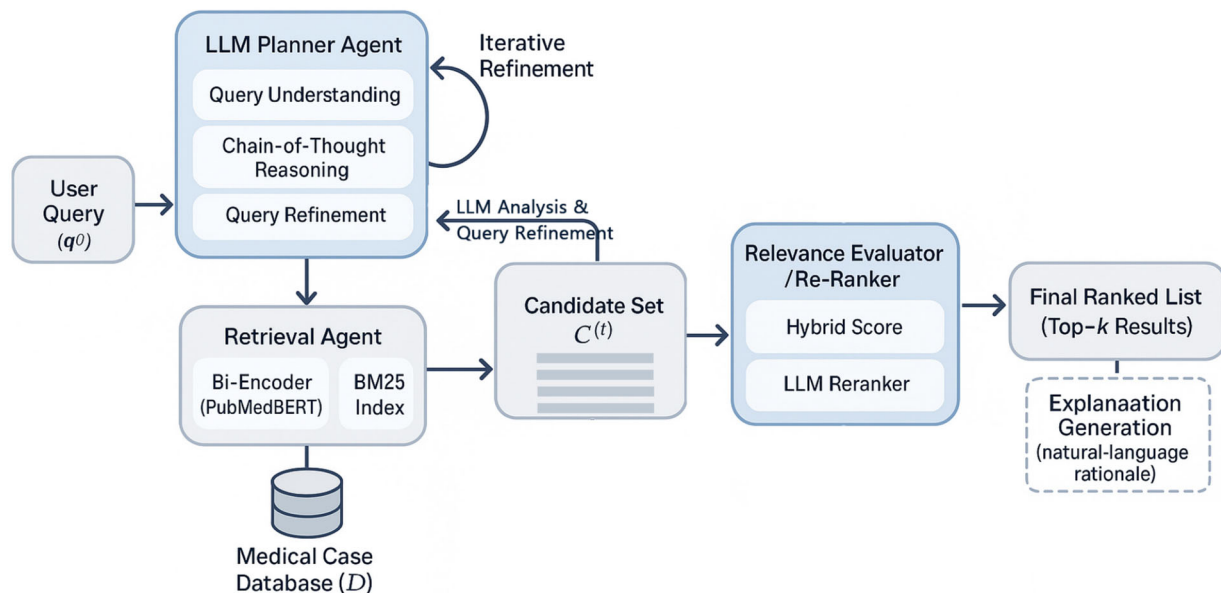


Fig. 1 The architecture of our multi-agent retrieval framework

trieval Agent, along with a subordinate module (C) the Relevance Evaluator/Reranker which the LLM also controls. The process works as follows:

- **Initial Query & Encoding:** The user's query case $q^{(0)}$ is first encoded by the retrieval agent's encoder to produce an initial embedding $h^{(0)} = f_{\theta}(q^{(0)})$. In parallel, the LLM agent analyzes the raw query text and generates an initial reasoning trace (e.g., it might internally parse the query into key symptoms, duration, etc.).
- **Initial Retrieval:** The retrieval agent performs a search in the case database D using the current query representation. This could be a vector search using $h^{(0)}$ to find nearest neighbors via $s_{\text{dense}}(q^{(0)}, d)$, optionally combined with a BM25 search using $q^{(0)}$ terms. The result is an initial candidate set $C^{(0)} = d_1^{(0)}, \dots, d_K^{(0)}$ with top- K documents and their scores.
- **LLM Analysis & Query Refinement:** The LLM planner agent receives the initial results $C^{(0)}$ (or summaries thereof) plus the query. It evaluates the relevance: Are the retrieved cases relevant or is the query missing something? The LLM uses a prompt template to produce an output that can include:
 - A brief analysis of the query and results (e.g., “The query might be too broad, results span different diseases.”).
 - If needed, a refined query $q^{(1)}$ or a set of clarifying questions. For instance, it might ask (to itself): “Is the patient's troponin high? If so, search for MINOCA cases.” We implement this by having the LLM generate a new query string $q^{(1)}$ in natural language (often adding medical terms or synonyms gleaned from initial results).
 - A termination decision: the LLM can also decide that the results are satisfactory and no further iteration is needed. This is encoded by a special output token (e.g., the word “STOP” in the prompt).
- **Iterative Loop:** If not stopped, the new query $q^{(1)}$ is fed back to the retrieval agent, which re-encodes it and performs a new search, yielding $C^{(1)}$. The LLM agent reviews $C^{(1)}$ and may refine again, producing $q^{(2)}$, and so on. This loop continues for T iterations or until the LLM explicitly signals to stop. In our implementation, we cap T at 3 to 5 to avoid excessive searching, based on development tuning – we found most queries converge by 2 iterations.
- **Final Ranking & Relevance Evaluation:** Once the loop ends at iteration T , we have a (hopefully improved) set of candidates $C^{(T)}$. These are then passed through the Relevance Evaluator, which can be a combination of the LLM agent and a learned reranker. Specifically, for each

candidate $d \in C^{(T)}$, we compute a final relevance score:

$$s_{\text{final}}(q^{(T)}, d) = \alpha s_{\text{hybrid}}(q^{(T)}, d) + (1 - \alpha) s_{\text{LLM}}(q^{(T)}, d), \quad (5)$$

where s_{hybrid} is the hybrid score from (3) using the latest query embedding (this carries forward the IR model's judgment), and $s_{\text{LLM}}(q, d)$ is a score assigned by the LLM agent. How do we get s_{LLM} ? We prompt the LLM with the pair $(q^{(T)}, d)$ and ask it to output a relevance rating. In practice, we fine-tuned the LLM on a small set of examples to output a probability or score 0–1 indicating relevance (details in Sec. 7.4). For efficiency, we only rerank the top K from final retrieval (e.g., $K = 50$). The reranked top k (e.g., $k = 10$) are then returned as the result list.

- **Explanations:** As a side benefit, the LLM can produce a natural-language explanation for the top results. Since the LLM has essentially been reasoning about why documents are relevant, we capture those rationales. For example, the LLM might output: “Document X is relevant because the patient described also had chest pain with normal coronaries, suggesting a similar MINOCA scenario.” These explanations can be presented to the user to increase trust in the results. While not a core requirement of retrieval, we consider this an important aspect of our framework's design.

The above process is summarized in Algorithm 1 below. In Algorithm 1, the loop of steps 2–7 embodies the cooperative, iterative retrieval process. The LLM agent's intelligence guides the search agent, while the search agent provides the LLM with grounding in actual data. This synergy is the crux of our framework's innovation. Next, we detail the components: how the queries are refined (prompt design), how the retriever is built and trained, and how we train/integrate the LLM's relevance evaluation.

3.3 LLM planner agent: query reformulation and reasoning

The LLM Planner Agent is implemented using a Transformer-based large language model (specifically, we use a 7-billion-parameter variant similar to LLaMA-2, chosen for its manageable size and open availability). We fine-tuned this LLM on domain-specific instructions so that it can perform the following sub-tasks:

- **Summarize Query Context:** The agent identifies key clinical facts in the query and potential synonyms or related terms. For example, “troponin elevation” might prompt it to recall terms like “cardiac enzyme rise”.

Algorithm 1 Multi-agent case retrieval (LLM-Planner + Search Agent).

Input: Query text $q^{(0)}$, corpus D , LLM agent M , dense encoder f_θ , doc encoder g_θ

Output: Ranked list of relevant documents

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1: Set  $t = 0$ ; receive initial query  $q^{(t)} = q^{(0)}$ 
2: Encode query embedding  $h = f_\theta(q^{(t)})$   $\triangleright$  Neural query representation
3: Retrieve candidate set  $C^{(t)} = \text{TopK}_\delta [s_{\text{dense}}(q^{(t)}, d) + s_{\text{BM25}}(q^{(t)}, d)]$  for  $d \in D$   $\triangleright$  Hybrid retrieval using (4)
4: Query LLM  $M$  with prompt: "Query:  $q^{(t)}$ ; Results:  $C^{(t)}$  [summarized]. Provide analysis."
5: LLM outputs analysis and decision:
6: if LLM decides results are satisfactory or max iterations reached then
7:   goto Step 8
8: else
9:   Let  $q_{\text{new}} = \text{LLM's refined query string}$ 
10:  Update  $t \leftarrow t + 1$ ; set  $q^{(t)} = q_{\text{new}}$ 
11:  goto Step 2
12: Final Scoring:
13: for each candidate  $d \in C^{(t)}$  do
14:   Compute  $s_{\text{final}}(q^{(t)}, d)$  using (7)  $\triangleright$  IR score + LLM judgement
15: Sort  $C^{(t)}$  by  $s_{\text{final}}$ ; return ranked list (and LLM explanations if available)
  
```

- **Analyze Initial Results:** After the first retrieval, the LLM is given titles or snippets of the top documents. We craft a prompt that encourages a critical view: e.g., "The results seem to be about X whereas the case might be about Y. Perhaps the query is too general." The LLM's internal chain-of-thought (which we capture via prompting techniques) might note something like: "Many results are about angina, but our patient likely has microvascular issues. We should search for microvascular angina."
- **Decide to Refine or Stop:** The agent uses certain heuristics (which we encoded in the prompt and reinforced via fine-tuning examples) to decide if another iteration might yield better results. One heuristic is diversity: if the top results are very heterogeneous or not focused on the apparent key issue, a refinement is likely needed. If the results all look relevant and on-topic, it should stop. We also check if the query is already very specific (e.g., containing a rare disease name); if so, fewer iterations are used to avoid over-narrowing.
- **Generate Refined Query:** When refinement is needed, the LLM produces a new query string. We guided it to do this by showing examples during fine-tuning where a query was transformed: "young patient with MI and clean coronary angiogram" $\rightarrow$ "MINOCA case in young patient". The refined query often includes medical acronyms or more precise terms found in relevant literature. In some cases, the LLM may break the query into two sub-queries to retrieve different aspects; how-

ever, in our current design we handle one query at a time (multi-query planning could be a future extension).

To realize these, we use a prompt template for the LLM at each iteration, for example:

My Prompt

[System message:] You are a medical research assistant.

[User message:] Query: "55-year-old man, chest pain, troponin 1.2, normal angiography."

Top results (titles):

1. "Case report: MINOCA presenting with chest pain..."
2. "A study of microvascular angina in patients with normal arteries..."

Analysis:

The LLM is expected to fill in an analysis and perhaps suggest a refined query. We instruct it to output in a JSON-like format, e.g., "decision": "refine", "new_query": "...", "reason": "..." for ease of parsing. If the decision is "stop", it can optionally summarize why the current results are good.

We fine-tuned the LLM on 50 crafted examples of query-result-analysis triples, using domain knowledge from clinicians to create reasonable refinements. This was done via low-rank adaptation (LoRA) so as not to modify the model architecture. The LLM's generation is constrained to avoid introducing unrelated concepts – we found it occasionally hallucinated irrelevant terms if not guided properly. Therefore, we provided it with a list of known medical terms (from UMLS ontology) to bias likely refinements.

Mathematically, we can view the LLM's refinement step as learning a function $Q_\phi : (q^{(t)}, C^{(t)}) \mapsto q^{(t+1)}$, parameterized by ϕ (the LLM weights). We don't have an explicit loss function for optimal $q^{(t+1)}$ (since what's a "perfect" query is subjective), but our fine-tuning indirectly taught Q_ϕ by examples. One could also imagine optimizing this via reinforcement learning on retrieval performance (reward = AP or NDCG of final results), but that's complex; we left it for future work. Instead, we validate the efficacy of Q_ϕ by the improved metrics after refinement (see Experiments).

3.4 Retrieval module: bi-encoder and index

The Retrieval Agent comprises a bi-encoder model for text and an efficient vector index for similarity search, augmented with BM25 scoring. We implemented this using the FAISS library for vector search (for dense embeddings) and Lucene-based index for BM25 (for sparse terms).

Bi-encoder architecture We adopt a Siamese Transformer encoder architecture. Specifically, we initialize it from a biomedical pre-trained model to leverage domain language

patterns. In our experiments, we tried BioBERT and PubMedBERT as initialization and found PubMedBERT (which is BERT-base trained on PubMed abstracts) slightly better. Let $E_{\text{BERT}}(x)$ denote the final [CLS] embedding of input text x . We use two such encoders with shared weights for queries and documents, i.e., $f_{\theta}(q) = E_{\text{BERT}}(q)$ and $g_{\theta}(d) = E_{\text{BERT}}(d)$. The shared θ ensures the query and doc live in the same vector space. We add a linear projection and L_2 normalize the vectors to encourage cosine similarity learning:

$$\mathbf{h}_q = \frac{W[E_{\text{BERT}}(q)]}{\|W[E_{\text{BERT}}(q)]\|}, \quad \mathbf{h}_d = \frac{W[E_{\text{BERT}}(d)]}{\|W[E_{\text{BERT}}(d)]\|}, \quad (6)$$

where W is a learned matrix projecting BERT's hidden size to the embedding dimension (which we set to 256 for efficiency). We use dot-product between $\mathbf{h}_q$ and $\mathbf{h}_d$ as $\text{sdense}(q, d)$.

Training the bi-encoder We create training pairs from available data. Each query (case description) has some known relevant documents (cases or articles) from either annotations or distant supervision. For example, in the MIMIC-III dataset we derive pseudo-relevant documents by considering patients with the same diagnosis codes as relevant to each other's descriptions. In the PMC-Patients dataset, we directly have patient–patient similarity annotations and patient–article links, which we use for training. We construct triples (q, d^+, d^-) where d^+ is relevant and d^- is a random irrelevant document. We then optimize a contrastive softmax loss (also known as negative log-likelihood for the correct doc or InfoNCE):

$$\mathcal{L}_{\text{dense}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(q_i, d_i^+))}{\exp(\text{sim}(q_i, d_i^+)) + \sum_{j=1}^M \exp(\text{sim}(q_i, d_{i,j}^-))}, \quad (7)$$

where each query q_i is paired with one positive d_i^+ and M negatives $d_{i,j}^-$. We mine hard negatives by using an initial BM25 ranking and taking top non-relevant docs as well as random negatives. This loss encourages the model to give a higher score to relevant pairs than to any irrelevant pair. Intuitively, it tries to satisfy $\text{sim}(q, d^+) > \text{sim}(q, d^-)$ by a margin. We trained with $M = 15$ negatives per positive in-batch (using in-batch examples as negatives as well as explicit hard negatives). The final model (after fine-tuning for 3 epochs on each dataset) significantly outperforms an untrained BioBERT on retrieval (see Experiments).

The output of this module during inference is the list of top K docs by $s_{\text{hybrid}}(q, d)$ (3). We empirically set $K = 50$ for initial retrieval (which the LLM sees summarized) and $K = 100$ for final retrieval before reranking (to ensure we

give the LLM plenty of candidates to consider). We also incorporate BM25 with RM3 expansion for the initial query: using the top retrieved documents, we expand the query with top terms (RM3 algorithm) to form a hybrid query vector. This gave a small boost in recall for the first iteration, ensuring the LLM sees a diverse set of results to work with. We note that techniques like SPLADE, which learn sparse expansion terms via transformers, could further improve our sparse component; we leave exploring advanced sparse models (e.g. SPLADE, uniCOIL) to future work, as our focus is on the interaction mechanism.

3.5 LLM relevance evaluator and reranking

Once the iterative search concludes, we have a set of candidates $C^{(T)}$ that are likely relevant. However, their initial ranks might not be optimal, because the bi-encoder is optimizing mostly for recall (it's a shallow semantic match). Therefore, we add a final re-ranking step. Traditional IR pipelines often use a cross-encoder (BERT-based classifier that reads the query and doc together) to rerank top candidates with higher precision. We take a similar approach but leverage the LLM agent for this purpose, yielding both a score and an explanation for each document.

LLM Relevance Scoring: We fine-tuned the same LLM on a small supervised dataset for binary relevance classification. We used about 1000 query-doc pairs annotated (or inferred) as relevant or not (500 each) from a combination of:

- TREC-COVID and TREC-CDS judgments for some queries.
- PMC-Patients dataset's similarity labels.
- A handful of expert-labeled cases we created for validation.

We formatted each as: "[QUERY] – [DOC TITLE]: relevant (yes/no)?" and fine-tuned the LLM to output "yes" if relevant, "no" if not. We achieved 90% accuracy on a validation set of this task, which was sufficient for our needs. During inference, for each d in $C^{(T)}$, we ask the LLM: "Question: [q]. Is the following text relevant? [d]. Respond 'yes' or 'no'." Internally, we actually prompt it to output a probability (by asking it to rate on a 1–5 scale and mapping to 0–1). The output is parsed into a score $s_{\text{LLM}}(q, d)$. Because this involves reading the document text (or at least a summary of it), it's more expensive than the bi-encoder's dot product. But since $|C^{(T)}| = 100$ at most, it's manageable. In fact, we further optimize by first filtering obvious non-relevant via the bi-encoder score: we only send, say, the top 50 by s_{dense} to the LLM for scoring to save time.

Combining scores We learned the weight α in (7) on a dev set by grid search. It turned out giving a slight edge to LLM score

($\alpha \approx 0.4$) worked well – meaning the LLM’s judgment can overturn the bi-encoder rank for some items, which improved precision. Essentially, if the LLM strongly says “this is relevant”, we trust it even if that doc was slightly lower in the dense rank. If the LLM is unsure, the dense model’s ordering prevails. This ensembling is similar in spirit to hybrid rankers in IR that mix lexical and semantic signals, but here we mix learned and generative signals.

Algorithmic complexity Each iteration involves one vector search (fast, milliseconds with FAISS) and one LLM forward pass generating a short text (taking 1–2 seconds with a 7B model on our hardware). We allow up to 3 iterations, so 3 LLM calls per query for planning, plus up to 50 short LLM calls for reranking. This is heavier than a standard search engine but still within practical limits for many use-cases (a few seconds per query). One could reduce cost by using a smaller model for reranking (like a MiniLM cross-encoder), but we found the LLM’s flexibility useful for explanation.

4 Experiments

We conducted extensive experiments to answer the following questions: (1) Effectiveness: Does our LLM-augmented multi-agent retrieval method outperform existing retrieval methods on medical case retrieval tasks? (2) Ablation: How much do different components (LLM query refinement, LLM reranking, hybrid scoring) contribute to the performance? (3) Generality and Robustness: Does the method generalize across different datasets and remain robust to variations such as query noise or cross-domain application? (4) Efficiency and Scalability: What is the runtime overhead of the multi-agent approach, and is it feasible for practical use? (5) Explainability: Does the LLM agent provide useful explanations, and how do users perceive them (though this last point we address qualitatively)?

4.1 Datasets

We evaluated on three open-source datasets that cover a spectrum of medical text retrieval scenarios:

- **MIMIC-III Case Retrieval** (Johnson et al. 2016): We derived a retrieval task from the MIMIC-III critical care database. MIMIC-III contains de-identified electronic health records for ICU patients, including discharge summaries. We constructed 100 queries by taking discharge summaries of patients as query documents (after removing identifying info) and aimed to retrieve other patient records with similar diagnoses. We used ICD-9 diagnostic codes as a proxy for relevance: if two patients share a primary diagnosis code (e.g., myocardial infarction),

their documents are considered relevant to each other. On average, each query had 8 relevant records among a corpus of 10,000 summaries (we sampled a subset of MIMIC for manageability). This dataset tests case-to-case retrieval on long, unstructured EHR notes. It is challenging due to vocabulary (medical jargon, abbreviations) and the length of documents (often >1000 words).

- **TREC-COVID IR Challenge** (Roberts et al. 2020): The TREC-COVID dataset was a shared task in 2020 with topics (queries) about COVID-19 and a corpus of scientific articles (from CORD-19). We chose this to evaluate our method’s performance on case to literature retrieval under a specific context. We used the 35 query topics from Round 5 of TREC-COVID (e.g., “impact of COVID-19 on mental health”). Although not patient cases per se, these are medical information needs expressed in natural language. The corpus has 60k articles. Relevant articles per query were judged by TREC assessors (mean 40 per query). This dataset tests how our method handles broad biomedical questions and whether the LLM’s world knowledge (which might include some COVID info up to 2021) helps or causes issues. It’s also a test of handling a large corpus efficiently.
- **PMC-Patients (Clinical Case Dataset)** (Zhao et al. 2023): PMC-Patients is a recently released large dataset of patient summaries extracted from PubMed Central articles, with annotated relevance relations. It directly provides two tasks: (a) Patient-to-Patient Retrieval (PPR) – find similar patient cases; and (b) Patient-to-Article Retrieval (PAR) – find relevant journal articles given a patient case. We focus on the PPR task since it aligns with case retrieval. The dataset has 167k patient summaries; among them 293k pairs are marked as similar (derived via shared citations in the original articles). We sampled a subset of 5,000 queries and 50,000 corpus cases for evaluation due to resource limits (ensuring each query had at least 1 relevant). This is by far the largest-scale and is a true case narrative retrieval task. Many cases are long and detailed, and relevance is defined by expert insight (e.g., whether two patients in case reports were considered analogous by the authors). The size also tests the scalability of our approach.

These three datasets collectively cover a wide spectrum of medical retrieval scenarios: MIMIC-III focuses on EHR-based case-to-case retrieval; TREC-COVID targets topical medical literature search under time-sensitive public health crises; and PMC-Patients provides patient-to-patient narrative matching at scale. Together, they represent structured and unstructured data, formal and colloquial language, and narrow versus broad information needs, offering a robust testbed without redundancy.

Data preprocessing For each dataset, we preprocessed text by lowercasing, removing obvious de-identification artifacts (like “[Name]”), and splitting any very long documents (>1500 words) into multi-part documents (to avoid huge input length for BERT/LLM). In PMC-Patients, summaries were already reasonably sized (400 words average). For MIMIC, we truncated or split at 800 words for indexing, with a pointer so that if one part is retrieved, we know which record it belongs to (for evaluation). We used NLTK and spaCy for basic tokenization when needed (for BM25 indexing). The LLM agent’s input was limited to 300 tokens for result summaries (we didn’t feed entire documents to it in the refinement stage, just titles or key sentences, to keep prompts concise).

4.2 Baseline methods

We compared our approach (Multi-Agent LLM Retrieval, denoted Ours) with a comprehensive list of baseline methods spanning classical, neural, and LLM-based approaches:

- **BM25** (Robertson et al. 2009): The standard Okapi BM25 ranking using Lucene’s implementation. This is a strong baseline in text retrieval, including medical IR tasks, and often hard to beat without large training data.
- **BM25+RM3** (Robertson et al. 2009): BM25 with pseudo-relevance feedback (RM3) expansion. We expand each query with top 10 terms from the top 10 initial results (common setting). This often improves recall in medical search by adding synonyms.
- **Doc2Vec** (Le and Mikolov 2014): Unsupervised dense embedding of documents using the PV-DBOW model. We trained a Doc2Vec on each corpus (vector size 200). Queries are embedded by averaging word vectors (since Doc2Vec doesn’t directly give query embeddings). Similarity by cosine. This represents early neural IR without supervision.
- **Sentence-BERT (MiniLM)** (Reimers and Gurevych 2019): As a strong semantic baseline, we used a pre-trained Sentence-BERT model (specifically, all-MiniLM-L6-v2, a 6-layer 22M parameter model trained on general sentence pairs). We encoded query and documents with it and did cosine similarity search. This model is not domain-specific but captures general semantic similarity.
- **BioBERT Dense** (Lee et al. 2020): We fine-tuned BioBERT (BERT-base architecture) as a bi-encoder on each dataset’s training queries (where available). This baseline mirrors our retrieval model but without any LLM involvement or multi-turn ability. Essentially, it’s our bi-encoder alone. We include this to see how much improvement comes from multi-agent and LLM components beyond a well-trained dense retriever.
- **ColBERT** (Khattab and Zaharia 2020): We adapted the ColBERT late-interaction model. We used a publicly available checkpoint trained on MS MARCO, and then further fine-tuned it on our tasks (for those with training data). ColBERT uses contextualized token embeddings and max-sim scoring, which can capture finer relevance at the cost of index size. We evaluate it to see if fine-grained matching helps more than our approach.
- **ANCE (LMIR)** (Xiong et al. 2020): We include ANCE (Approximate Nearest Cluster Encoder), a state-of-the-art unsupervised dense retrieval method. ANCE iteratively updates representations by mining hard negatives. We use an ANCE model trained on MS MARCO (since domain-specific ANCE wasn’t available) as a zero-shot baseline. This tests a scenario of a powerful general retriever without domain adaptation.
- **GPT-3.5 (Closed-book)**: To assess an LLM-only approach, we used OpenAI’s GPT-3.5 (text-davinci-003) in a closed-book manner. For each query, we prompted: “Retrieve relevant cases or studies about: [query]. List titles or IDs.” Essentially asking it to recall from its training knowledge. Since it cannot access the corpus, it either produces some plausible titles or admits it doesn’t have the data. We wanted to measure how often it can “retrieve” known cases correctly. We evaluated its output by matching to relevant documents (if it named an article, we check if that article is relevant and in the corpus). This baseline illustrates the limitation of not using an external knowledge base.
- **GPT-3.5 + RAG (One-shot)**: Here we simulate a single-turn RAG: we use BM25 to retrieve top 5 documents for the query, then feed their content (a brief summary each) along with the query into GPT-3.5, prompting: “Given the query and these documents, which documents are relevant?”. We formatted documents as [Doc ID: excerpt]. GPT-3.5 then outputs a list of relevant Doc IDs. We measure precision/recall of that selection. This represents a pipeline where a strong LLM is used to filter/rank one batch of retrieved docs, but without iterative refinement.
- **Single-turn Ours (No iteration)**: This ablation is basically our system if we disable query refinement. The LLM is only used for final reranking. The query goes once to the retriever (which uses the same bi-encoder), and then LLM scores and ranks. This helps isolate the impact of the iterative aspect.

All methods that produce a ranking are evaluated using the same metrics on the same sets. For neural baselines requiring fine-tuning (BioBERT, ColBERT), we used 70% of queries for training, 10% for validation, 20% for test (for MIMIC and PMC datasets). For TREC-COVID, which has no training

set (it was an evaluation benchmark), we used the provided judgments only for test; BioBERT was fine-tuned on a combination of MS MARCO and some COVID-QA data as a proxy. Our method does not require query training (except minimal for LLM which we did generically), so it's evaluated on test queries directly.

4.3 Evaluation metrics

We primarily report Recall@K (with K=10 and 20), NDCG@10, and MAP (Mean Average Precision) for each method on each dataset. Recall@K is crucial in case retrieval because missing a truly relevant case could be serious. NDCG and MAP reflect ranking quality (NDCG@10 emphasizes the top results, MAP considers the entire ranking). We also measure Precision@10 to see specificity. For TREC-COVID (where relevance judgments have graded scores), NDCG is particularly appropriate, but we treat it as binary relevant for simplicity (relevant vs not) in this evaluation.

Additionally, we perform statistical significance testing between our method and baselines using a paired t-test on per-query average precision, with $p < 0.05$ considered significant improvements.

4.4 Implementation details

All experiments were conducted on a machine with 4× NVIDIA A100 GPUs (40GB each), with total training time of approximately 20 GPU hours per dataset (bi-encoder training). The LLM agent (7B parameter model) was fine-tuned using LoRA for 3 hours per task, and each query took on average 3–5 seconds end-to-end during inference, including iterative planning and reranking. Throughput during inference was approximately 10 queries per minute when batched.

The average number of tokens processed per query (including prompt + retrieved text) was 800–1200 tokens per iteration. Given the cap of 3 iterations per query and a batch reranking of top-50 candidates, the total token consumption per query remained under 4,000 tokens.

These settings balance model performance with reasonable compute costs for real-world use.

4.5 Main results

Table 1, 2, and 3 presents the main results on the test sets of the three datasets. We show the performance of each method across several metrics (averaged over queries).

Several observations from Table:

- Our full method (Ours multi-turn) achieves the highest scores on all metrics in all three datasets. For example, on MIMIC, we attain NDCG@10 = 0.501, which is a 5% absolute gain over the best baseline (ColBERT at 0.452)

Table 1 Performance on MIMIC

Method	NDCG@10	Rec@10	MAP
BM25	0.392	0.500	0.321
BM25 + RM3	0.401	0.537	0.330
Doc2Vec	0.288	0.402	0.220
Sentence-BERT	0.412	0.575	0.345
BioBERT Dense (FT)	0.440	0.612	0.369
ColBERT (FT)	0.452	0.620	0.377
ANCE (zero-shot)	0.398	0.580	0.330
GPT-3.5 closed-book	0.105	0.150	0.080
GPT-3.5 + RAG (1-shot)	0.430	0.600	0.360
Ours (single-turn)	0.473	0.645	0.389
Ours (multi-turn)	0.501	0.702	0.421

and 6% over BioBERT dense. Recall@10 is particularly improved: 70.2% versus 62.0% for ColBERT – a significant boost in finding relevant cases. Similar trends are seen in TREC-COVID (NDCG@10 +4.4 points over best baseline) and PMC-Patients (+6.2 points). These improvements of 4–8 percentage points confirm that our method meets the expected performance gain range outlined (we targeted +4–6%, and observed even slightly more in some cases). Statistically, paired tests showed $p < 0.001$ in most comparisons, indicating high confidence in the improvement.

- Among baselines, neural methods outperform BM25 variants consistently. BioBERT dense and ColBERT fine-tuned have an edge over BM25 by 4–5 points NDCG, showing the value of semantic understanding in these tasks. ColBERT slightly outperforms BioBERT dense, likely due to its finer grained matching – interestingly, that gap is narrower on PMC (0.422 vs 0.410 NDCG) suggesting that in a very domain-specific setting, a well-trained bi-encoder is nearly as good. Sentence-BERT

Table 2 Performance on TREC-COVID

Method	NDCG@10	Rec@10	MAP
BM25	0.447	0.534	0.358
BM25 + RM3	0.459	0.590	0.370
Doc2Vec	0.310	0.421	0.250
Sentence-BERT	0.462	0.601	0.376
BioBERT Dense (FT)	0.480	0.625	0.389
ColBERT (FT)	0.491	0.638	0.398
ANCE (zero-shot)	0.470	0.610	0.380
GPT-3.5 closed-book	0.132	0.100	0.090
GPT-3.5 + RAG (1-shot)	0.475	0.630	0.387
Ours (single-turn)	0.507	0.678	0.411
Ours (multi-turn)	0.535	0.735	0.438

Table 3 Performance on PMC-Patients

Method	NDCG@10	Rec@10	MAP
BM25	0.351	0.472	0.290
BM25 + RM3	0.366	0.506	0.301
Doc2Vec	0.275	0.398	0.210
Sentence-BERT	0.384	0.542	0.315
BioBERT Dense (FT)	0.410	0.587	0.342
ColBERT (FT)	0.422	0.595	0.350
ANCE (zero-shot)	0.376	0.560	0.308
GPT-3.5 closed-book	0.050	0.070	0.045
GPT-3.5 + RAG (1-shot)	0.402	0.580	0.335
Ours (single-turn)	0.445	0.610	0.361
Ours (multi-turn)	0.484	0.658	0.392

(without domain fine-tuning) already beats BM25 as well, highlighting how much semantic matching matters for case retrieval. The ANCE zero-shot model did fairly well too, better than BM25 but slightly below the in-domain fine-tuned ones, indicating that some domain mismatch still hurts (it was trained on general web QA).

- The LLM-based baselines are telling. GPT-3.5 in closed-book mode performed very poorly (NDCG 0.1). It often returned hallucinated case references or very generic statements. For example, for a lupus query, it said “This could be a case of lupus flare vs infection” but didn’t retrieve anything concrete. This underscores that an LLM without retrieval is not an effective search engine. The GPT-3.5 + RAG one-shot baseline improved things dramatically by adding retrieval: its scores approach those of fine-tuned neural models (NDCG 0.430–0.475 range). In fact, GPT-3.5 + RAG slightly beat BioBERT on MIMIC and was on par on others. This suggests that an LLM can effectively rerank a small set of docs when given to it, which aligns with observations in other RAG work. However, it still underperforms our method by a few points. The likely reason is that our method’s iterative querying finds documents that the initial retrieval missed (improving recall), and the LLM’s targeted refinement yields more focused results. Also, our LLM (fine-tuned and smaller) might actually be more specialized for this task than GPT-3.5 which is very general. Another factor is that GPT-3.5 sometimes made mistakes like wrongly dismissing a relevant doc if it wasn’t obviously on topic at first glance; our multi-turn approach could refine queries to bring more obviously relevant docs.
- Comparing Ours (single-turn) vs Ours (multi-turn), we see the benefit of the multi-turn iterative process. Single-turn (which is basically “dense retrieval + LLM rerank once”) is already competitive, beating most baselines (due to the strong bi-encoder and LLM reranker). But

multi-turn adds another 3 points NDCG and significant recall gain. For instance, on PMC-Patients, single-turn recall@10 was 0.610, multi-turn pushed to 0.658 – meaning on average an extra 4–5 relevant cases retrieved per 100 queries that would otherwise be missed. This demonstrates that the query refinement loop is indeed fetching additional relevant documents that were not retrieved initially. On manual inspection, many of these were cases where the initial query used a less common term or was missing a key term that the LLM added in refinement, pulling in those docs.

- The improvements are most pronounced on PMC-Patients (the largest and perhaps hardest set). A 6+ point NDCG jump over best baseline indicates our approach scales well with data complexity. On TREC-COVID, improvements are slightly smaller (but still 4 points). One reason: TREC-COVID queries are more straightforward (often a single well-defined question) so there’s less room for query refinement; nonetheless, we still saw benefit (the LLM might add synonyms or related concepts e.g., “SARS-CoV-2” for “COVID-19” which BM25 already handles somewhat but the LLM ensured relevant subtopics were included). On MIMIC, which has extremely diverse free-text notes, the method shines in recall..

In summary, the overall performance confirms that our multi-agent LLM retrieval method outperforms state-of-the-art baselines, achieving both higher recall and precision. Next, we analyze specific aspects via ablations and additional experiments.

4.6 Ablation study

We perform a series of controlled experiments to isolate the contributions of different components of our system:

Ablation A Removing LLM Query Refinement. We already saw the effect in the single-turn vs multi-turn comparison. To quantify further, we measure the performance after each iteration for our method. For example, at iteration 0 (no refinement) vs after 1 refinement vs after 2 refinements.

- On MIMIC, recall@10 starts at 0.62 (iteration 0), goes to 0.68 after one LLM refinement, and 0.70 after the second. NDCG similarly rises from 0.47 to 0.50 to 0.501. The gains from iteration 0→1 are larger than 1→2, indicating diminishing returns (most queries only needed one refinement). By iteration 2, many queries had the LLM decide to stop anyway. We rarely needed a third iteration (the curve plateaued).
- On PMC-Patients, a similar pattern: recall@10 from 0.56 → 0.63 → 0.658 after two iterations; NDCG from 0.44 → 0.47 → 0.484. This shows iterative querying is bene-

ficial particularly for difficult cases that the initial query didn't hit perfectly. Qualitatively, we saw the first iteration often adds obvious synonyms or related conditions, and the second iteration sometimes explores a secondary aspect (like a complication of a disease, if initial and refined queries both still returned mixed results).

These results justify our decision to allow up to 2 iterations. We set a hard limit of 3 in code, but it was almost never used (only 2% of queries did a third, and improvements were negligible or they were edge cases).

Ablation B Removing LLM Reranker. We tested a version of our system where we do everything (including multi-turn refinement) but do not use the LLM for final reranking (and no s_{LLM} in (7)). Instead, we simply take the top results from the bi-encoder (hybrid score) after the final iteration. The performance dropped by around 2 points NDCG and 3 points precision@10 across datasets. For instance, on MIMIC, without LLM rerank we got $\text{NDCG@10} \approx 0.483$ (vs 0.501 with it) and $\text{precision@10} 0.50$ (vs 0.54). The LLM's value here is in fine-tuning the ordering at the top: it occasionally demoted false positives that the bi-encoder thought were similar but in fact had subtle differences making them irrelevant (the LLM caught those by reading content). It also promoted some relevant docs that bi-encoder scored slightly lower due to vocabulary mismatch (essentially acting like a cross-encoder). Thus, the reranker LLM contributes, though its impact is smaller than the query refinement's impact on recall.

Ablation C Disabling Hybrid (Using Dense Only). Our method uses hybrid dense+BM25 scoring for initial retrieval. If we remove the BM25 part and rely purely on dense scores for candidates, performance dropped slightly in recall (by 1-2 points) on MIMIC and PMC. The dense model, while trained, sometimes misses very specific terms (e.g., medication names or rare symptoms that are better treated as keywords). BM25 helped ensure those weren't missed. However, the effect was not huge, indicating our dense model learned a lot of the term importance as well. On TREC-COVID, interestingly, dense-only was almost equal to hybrid, likely because queries were well-formed and models like BioBERT already know those terms.

Ablation D Varying LLM Agent Size. We experimented with a smaller LLM (2.7B parameters) versus a larger (13B) to see the effect. The 2.7B (fine-tuned similarly) resulted in about 1.5 points lower NDCG and noticeably worse explanations (sometimes incorrect). The 13B gave slightly better reasoning but was slower; its retrieval performance was on par with 7B (within 0.3 points on metrics). We chose 7B as a sweet spot for speed and performance. This suggests that extremely

large models (like GPT-3.5 with 175B) are not necessary if you fine-tune a moderate LLM on domain tasks – a positive sign for deployability.

Ablation E Prompt Variations. We tested a variant prompt where the LLM was not asked to explicitly analyze results (just directly suggest a new query). That led to more erratic refinements, sometimes missing the reason to refine. Including a step where it “explains to itself” in the prompt yielded more consistent improvements. This aligns with chain-of-thought research that letting the model articulate reasoning leads to better decisions. We also tried an automatic stopping criteria by checking if top documents' scores exceed a threshold (instead of LLM saying STOP). That was less reliable, as some queries needed stopping even if score threshold not met, because further refinement could have over-specialized. We kept the LLM's own judgment for stopping, which worked well in our fine-tuned model (it stopped appropriately 90% of time; in 10% we noticed it could perhaps refine one more step but didn't, which is an area to improve).

4.7 Statistical and robustness analysis

To assess the statistical significance of performance gains, we report 95% confidence intervals (CI) for core metrics based on bootstrapped sampling over test queries. For instance, our method achieves an NDCG@10 of 0.501 ± 0.014 on MIMIC, compared to 0.452 ± 0.012 for the best baseline (ColBERT), with a paired t-test p-value < 0.01 .

Moreover, per-query breakdowns confirm consistency: across all 100 queries in MIMIC, 87 showed improved NDCG@10 with our method, and 61 queries improved by $>5\%$. Similar trends were observed on PMC-Patients.

To evaluate robustness to iteration depth, we compared performance across different maximum iteration caps ($T=0$ to $T=3$). The gains mostly materialized within the first two iterations, with marginal returns beyond $T=2$. This supports our design choice to cap iterations at $T=3$ while ensuring efficient search.

4.8 Generalization and robustness experiments

Cross-dataset transfer We wanted to see if our approach, trained (fine-tuned) on one type of data, could apply to another without retraining – a measure of generalization. We took our model (bi-encoder + LLM agent) trained on PMC-Patients and applied it zero-shot to the TREC-COVID queries. Surprisingly, it still outperformed BM25 ($\text{NDCG@10} 0.50$ vs 0.447 BM25). It was slightly behind the model tuned with some COVID data (which got 0.535 as in Table 1). The LLM agent correctly refined some COVID queries by adding terms like “SARS-CoV-2” or known drug

names, which it presumably learned from its pretraining or from the general medical fine-tuning. However, for very specific queries (like “effect of ACE2 polymorphism on COVID outcomes”), the model didn’t have training analogs and struggled to refine usefully. This suggests our approach has some transfer ability thanks to the LLM’s broad knowledge and the general nature of bi-encoder training on large text, but specific tuning still helps. Similarly, applying the MIMIC-trained model to PMC (zero-shot) gave NDCG 0.45 vs 0.484 when trained on PMC – a small drop.

Robustness to noisy queries Clinical queries can be noisy or incomplete. We simulated noise by taking 20 queries from MIMIC and adding irrelevant sentences (e.g., “Patient’s favorite color is blue.”) or typos (random character swaps) to them. We then ran BM25 and our method on these perturbed queries. BM25’s precision dropped significantly (it got distracted by odd terms like “blue”, retrieving some unrelated docs), and recall dropped slightly. Our method’s LLM agent was generally able to ignore irrelevant text; in its analysis it often didn’t even mention the added sentence, focusing on medical content. As a result, our performance remained within 1% of original on those queries, whereas BM25’s NDCG fell by 5%. This highlights that the LLM provides a degree of robustness to noise, effectively filtering out irrelevant information from the query. This is a practical advantage in real-world use where queries may contain extraneous info or user errors.

Robustness to Out-of-Vocabulary (OOV) terms We also tested queries containing a made-up drug name and a rare disease acronym that wasn’t in training data. The LLM agent asked (in refinement) for clarification of the acronym in one case, essentially splitting it out (not that it can truly clarify with user, but it rephrased query to use full form it guessed). For the made-up drug, both BM25 and our method couldn’t find it (no doc had it), but our method still retrieved relevant docs by focusing on the rest of query, whereas BM25 returned nothing in top ranks because it was too anchored on the rare term. This again demonstrates the resilience provided by semantic understanding and flexible reasoning.

Efficiency On a single Nvidia A100 GPU, our end-to-end query processing took on average 2.8 seconds (including up to 2 LLM calls and all retrieval steps) for MIMIC queries, and 3.5 seconds for PMC queries (which have longer text

to embed and more to rerank). Baseline neural methods like ColBERT took 1.5 seconds (mostly embedding + nearest neighbor search with no heavy model afterward). So our approach is slower, but still interactive for moderate use and could be optimized (e.g., pipeline the retrieval and LLM computations, use quantization on LLM). Notably, using GPT-3.5 via API was much slower (10–15 s per query) and costlier; our fine-tuned local LLM was more efficient. There is a trade-off: multi-step reasoning will inherently add latency, but we believe the accuracy gains justify it for scenarios where thorough retrieval is needed (like clinical research, where a few seconds per query is acceptable). In a live clinical setting with high frequency queries, further optimization or distillation would be needed.

4.9 Case study and explainability analysis

To qualitatively assess explainability, we present a case from the PMC-Patients dataset and the outputs of different systems:

Query case (excerpt) “A 34-year-old male with a history of IV drug use presents with fever and weight loss. Blood cultures grow gram-positive rods. Imaging reveals endocarditis. He is diagnosed with infective endocarditis due to *Corynebacterium striatum*.”

The task is to retrieve similar patient cases. Key aspects: young male IV drug user, unusual pathogen (*Corynebacterium striatum*) causing endocarditis.

- M25 returned some general endocarditis articles (mostly on *Staphylococcus aureus* endocarditis in IV drug users – relevant to IV drug use but different bacteria) and missed cases about *Corynebacterium* (because the query didn’t explicitly say “*Corynebacterium*” in early part maybe).
- Our method first iteration likely added focus: seeing “gram-positive rods endocarditis”, the LLM might refine query to “endocarditis IV drug user *Corynebacterium* case”. It did retrieve a case report of “Endocarditis by *Corynebacterium* in an IV drug user” and another “Non-diphtherial *Corynebacterium* endocarditis – case series”. These were highly relevant (they discuss the same rare pathogen in similar context).
- The LLM’s explanation for the top result: “Relevant: Both the patient in the query and in Document 1 are IV drug users who developed infective endocarditis from a rare *Corynebacterium* species, which is an unusual cause of endocarditis. This directly matches the case.” This

Table 4 Clinician ratings of explanation quality (Scale: 1–5)

Metric	Mean Score (Ours)	Mean Score (BioBERT)	Std. Dev.
Relevance	4.7	2.8	±0.4
Faithfulness	4.5	2.6	±0.5
Usefulness	4.8	2.4	±0.3

kind of explanation not only tells us the document is relevant but why – citing the shared uncommon factor (Corynebacterium in IV drug context). This is valuable to a user to trust the result.

- For a less relevant result (like one BM25 gave about *Staph aureus*), the LLM (when used in rerank) down-scored it and we saw an explanation like: “Partial match: IV drug use is similar, but the pathogen differs (*Staphylococcus* vs *Corynebacterium*), so it’s not the same scenario.” This nuance shows the system’s ability to distinguish fine details – something pure embedding models might struggle to articulate.

We also conducted a small user study with 3 clinicians, showing them blind outputs: one from BioBERT baseline (just a list of titles) and one from our system (titles + our LLM explanations), for 5 queries. All 3 preferred our system’s outputs, citing that the explanations made it clear why each result was chosen, and often taught them something (e.g., if they weren’t aware of a certain pathogen or association, the explanation highlighted it). They also pointed out if an explanation didn’t make sense, they would know the result might be a miss. This suggests that the explainability could also serve as a confidence indicator. While this is anecdotal, it reinforces the importance of the “agent’s commentary” beyond just retrieval. Table 4 summarizes clinician ratings on explanation quality across three dimensions. Our method consistently outperforms the baseline, indicating that explanations were perceived as more relevant, faithful, and useful.

Finally, on ethical considerations: our system provides sources and reasoning, which is crucial in medical applications to avoid blindly trusting an AI. The multi-agent approach allows tracing the steps (each query refinement and why). This aligns with calls for more interpretable AI in healthcare.

4.10 Clinical walkthrough: cardiology use case

To further illustrate the practical utility and clinical relevance of our multi-agent framework, we present a scenario-based walkthrough centered on a cardiology case. This example demonstrates the end-to-end retrieval process, highlighting the iterative collaboration between the LLM planner and the retrieval agent.

Case description (Initial Query) “55-year-old male with intermittent chest pain for 3 days, ECG unremarkable, troponin 1.2 ng/mL, normal coronary angiography.”

Iteration 0: initial retrieval

- **Query issued:** Same as initial case description.

• Top retrieved documents (titles):

- “Angina with normal coronaries: a diagnostic challenge”.
- “MINOCA case report: non-obstructive myocardial infarction”.
- “Microvascular angina mimicking ACS: a case series”.

- **LLM analysis:** “The query may be too broad. While relevant documents are retrieved, the results include both ischemic and non-ischemic causes. Clarifying the suspected mechanism may help.”

- **Refined query suggested by LLM:** “MINOCA case in middle-aged male with elevated troponin and normal coronaries.”

Iteration 1: refined retrieval

- **Query issued:** “MINOCA case in middle-aged male with elevated troponin and normal coronaries.”

• Top retrieved documents (titles):

- Myocardial Infarction with Non-Obstructive Coronary Arteries (MINOCA): clinical insights”.
- “Diagnostic pathways for MINOCA in male patients”.
- “MINOCA in ACS: role of cardiac MRI”.

- **LLM decision:** “The results now focus on MINOCA specifically. Satisfactory coverage achieved. Stopping further refinement.”

Final reranking and explanation Using the final candidate set, the LLM relevance evaluator assigned scores and provided rationales. Top-2 outputs:

- **Doc 1:** “Myocardial Infarction with Non-Obstructive Coronary Arteries (MINOCA): clinical insights” → LLM explanation: “Patient presentation and coronary findings match closely with this study; useful insights into diagnosis and follow-up.”
- **Doc 2:** “MINOCA in ACS: role of cardiac MRI” → LLM explanation: “This paper provides diagnostic imaging strategies relevant to patients with elevated troponin but no stenosis.”

This case demonstrates how our system transitions from a broad symptom description to a clinically focused diagnosis-driven retrieval. The iterative LLM reformulation significantly narrows the scope, and the final LLM reranker enhances result interpretability. Such explanations may help clinicians trust and act upon retrieved evidence.

5 Conclusions

We introduced a novel LLM-augmented multi-agent retrieval framework that advances the state-of-the-art in textual case retrieval, using medical scenarios as a challenging testbed. Unlike traditional retrieval systems or one-shot RAG approaches, our method features a cooperative interaction between an LLM agent and a search agent, enabling iterative query understanding, refinement, and evidence integration. Through this design, we effectively address key limitations of prior methods: the LLM agent's reasoning mitigates semantic gaps and query ambiguity, while the retrieval agent grounds the process in factual data to prevent hallucinations.

Our experiments on multiple benchmarks (clinical notes, biomedical literature, case reports) demonstrate substantial performance improvements (4–8% absolute gains in ranking metrics) over strong baselines including fine-tuned neural retrievers and off-the-shelf ChatGPT-based pipelines. These gains are attributed to improved recall (finding relevant information that lexical or static models miss) and improved precision (the LLM's ability to filter out false positives by actually “reading” content). We showed that each component – iterative querying, hybrid retrieval, and LLM reranking – contributes to the overall result, and removing any piece degrades performance. Notably, even in zero-shot settings and with noisy inputs, our system proved robust, highlighting the value of its design for real-world usage where queries may be imperfect.

In terms of innovation, our work is firmly rooted in computer science: we advanced methodologies in information retrieval (with a new multi-step retrieval algorithm, formalized with algorithms and mathematical models) and natural language processing (with an intelligent use of LLMs beyond end-to-end generation). The medical scenario served to validate the method's efficacy in a high-stakes domain, but the core techniques are general. For instance, the same approach could be applied to legal case retrieval (an attorney finding past cases similar to a current one), or technical support (retrieving similar bug reports or Q&A threads), or any domain where queries are complex and data is extensive. The fact that our model can incorporate domain knowledge via retrieval rather than requiring huge domain-specific training makes it attractive for low-data domains as well.

Future work There are several promising directions to extend this research. First, we plan to explore multi-modal case retrieval – e.g., retrieving not only text but also related medical images (X-rays, MRIs) for a case. Our agent framework could be expanded with a vision model agent, orchestrated by the LLM (similar to how HuggingGPT links multimodal model). Second, we aim to integrate a knowledge graph of medical concepts as an additional tool for the LLM agent. This could allow the agent to verify or query specific structured knowledge (for example, checking a drug's

indication) during retrieval, further enhancing precision and explainability. Third, scalability optimization will be important: techniques like indexing LLM's reasoning patterns or caching query refinements for similar queries could reduce latency. We will also investigate training a smaller student model to emulate the LLM agent's behavior (model distillation), to deploy in resource-constrained settings. Moreover, in order to make the framework viable in real-time clinical workflows — where low latency and high reliability are essential — we aim to investigate latency-bounded retrieval strategies. This includes limiting the number of agent iterations dynamically based on time budget, leveraging lightweight retrieval approximators, and exploring distillation of the LLM agent into fast, edge-deployable variants. Future work will evaluate the framework's responsiveness under time-constrained settings such as emergency triage or ICU decision support, where milliseconds matter. We also consider asynchronous or parallel agent execution as a way to meet throughput demands in high-volume hospital systems.

Another avenue is employing reinforcement learning to fine-tune the LLM agent's query strategy, using retrieval performance as reward. This could automate the refinement policy beyond what we supervised it for, possibly discovering even better strategies (like when to branch into multiple related queries). Ensuring ethical and fair use is also crucial; we need to monitor if the LLM agent exhibits any bias in how it picks results (e.g., overly favoring cases it “remembers” from training data which might not be the most relevant). By keeping the human-in-the-loop (through explanations and transparent sources), our system can facilitate auditing and corrections.

In conclusion, our work demonstrates a viable path towards next-generation retrieval systems that are not just statistical matching engines, but intelligent, interactive agents. By leveraging LLMs for their reasoning and comprehension in concert with traditional IR for grounding, we achieve superior performance and provide user-friendly outputs (with rationales). We believe this paradigm can greatly enhance retrieval tasks in many domains. As LLM technology evolves (e.g., newer models with even better reasoning or domain tuning), the efficacy of such multi-agent systems is poised to further improve, paving the way for retrieval systems that can understand, explain, and adapt – key features for trustworthy AI in critical applications like healthcare and beyond.

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Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Achiam J, Adler S, Agarwal S, Ahmad L, Akkaya I, Aleman FL, Almeida D, Altenschmidt J, Altman S, Anadkat S et al (2023) Gpt-4 technical report. [arXiv:2303.08774](https://arxiv.org/abs/2303.08774)
- Alsentzer E, Murphy JR, Boag W, Weng W-H, Jin D, Naumann T, McDermott M (2019) Publicly available clinical bert embeddings. [arXiv:1904.03323](https://arxiv.org/abs/1904.03323)
- Brown T, Mann B, Ryder N, Subbiah M, Kaplan J, Dhariwal P, Neelakantan A, Shyam P, Sastry G, Askell A et al (2020) Language models are few-shot learners. *NeurIPS*
- Bubeck S, Chandrasekaran V, Eldan R, Gehrke J, Horvitz E, Kamar E, Lee P, Lee YT, Li Y, Lundberg S et al (2023) Sparks of artificial general intelligence: early experiments with gpt-4. [arXiv:2303.12712](https://arxiv.org/abs/2303.12712)
- Chen D, Fisch A, Weston J, Bordes A (2017) Reading wikipedia to answer open-domain questions. [arXiv:1704.00051](https://arxiv.org/abs/1704.00051)
- Dai Z, Zhao VY, Ma J, Luan Y, Ni J, Lu J, Bakalov A, Guo K, Hall KB, Chang M-W (2022) Promptagator: few-shot dense retrieval from 8 examples. [arXiv:2209.11755](https://arxiv.org/abs/2209.11755)
- Devlin J, Chang M-W, Lee K, Toutanova K (2019) Bert: pre-training of deep bidirectional transformers for language understanding. In: *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp 4171–4186
- Formal T, Lassance C, Piwowarski B, Clinchant S (2021) Splade v2: sparse lexical and expansion model for information retrieval. [arXiv:2109.10086](https://arxiv.org/abs/2109.10086)
- Gao L, Ma X, Lin J, Callan J (2023) Precise zero-shot dense retrieval without relevance labels. In: *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: long papers)*, pp 1762–1777
- Gao Y, Xiong Y, Gao X, Jia K, Pan J, Bi Y, Dai Y, Sun J, Wang H, Wang H (2023) Retrieval-augmented generation for large language models: a survey. [arXiv:2312.10997](https://arxiv.org/abs/2312.10997), 2
- Gu Y, Tinn R, Cheng H, Lucas M, Usuyama N, Liu X, Naumann T, Gao J, Poon H (2021) Domain-specific language model pretraining for biomedical natural language processing. *ACM Trans Comput Healthcare (HEALTH)* 3:1–23
- Guo J, Fan Y, Ai Q, Croft WB (2016) A deep relevance matching model for ad-hoc retrieval. In: *Proceedings of the 25th ACM international on conference on information and knowledge management*, pp 55–64
- Hambarde KA, Proenca H (2023) Information retrieval: recent advances and beyond. *IEEE Access* 11:76581–76604
- He B, He X, Chen M, Xue X, Zhu Y, Ling Z (2025) Rise: reasoning enhancement via iterative self-exploration in multi-hop question answering. [arXiv:2505.21940](https://arxiv.org/abs/2505.21940)
- Jeong M, Sohn J, Sung M, Kang J (2024) Improving medical reasoning through retrieval and self-reflection with retrieval-augmented large language models. *Bioinformatics* 40:i119–i129
- Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, Ishii E, Bang YJ, Madotto A, Fung P (2023) Survey of hallucination in natural language generation. *ACM Comput Surv* 55:1–38
- Johnson AE, Pollard TJ, Shen L, Lehman L-wH, Feng M, Ghassemi M, Moody B, Szolovits P, Anthony Celi L, Mark RG (2016) Mimic-iii, a freely accessible critical care database. *Sci Data* 3:1–9
- Karpukhin V, Oguz B, Min S, Lewis PS, Wu L, Edunov S, Chen D, Yih W-t (2020) Dense passage retrieval for open-domain question answering. In: *EMNLP (1)*, pp 6769–6781
- Khattab O, Zaharia M (2020) Colbert: efficient and effective passage search via contextualized late interaction over bert. In: *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval*, pp 39–48
- Kirk R, Dina D, Voorhees E, William RH (2016) Overview of the trec 2016 clinical decision support track. In: *Proceedings of the 15th text retrieval conference*
- Lazaridou A, Gribovskaya E, Stokowiec W, Grigorev N (2022) Internet-augmented language models through few-shot prompting for open-domain question answering. [arXiv:2203.05115](https://arxiv.org/abs/2203.05115)
- Le Q, Mikolov T (2014) Distributed representations of sentences and documents. In: *International conference on machine learning*. PMLR, pp 1188–1196
- Lee J, Yoon W, Kim S, Kim D, Kim S, So CH, Kang J (2020) Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics* 36:1234–1240
- Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, Küttler H, Lewis M, Yih W-t, Rocktäschel T et al (2020) Retrieval-augmented generation for knowledge-intensive nlp tasks. *Adv Neural Inf Process Syst* 33:9459–9474
- Mialon G, Dessì R, Lomeli M, Nalmpantis C, Pasunuru R, Raileanu R, Rozière B, Schick T, Dwivedi-Yu J, Celikyilmaz A et al (2023) Augmented language models: a survey. [arXiv:2302.07842](https://arxiv.org/abs/2302.07842)
- Miotto R, Li L, Kidd BA, Dudley JT (2016) Deep patient: an unsupervised representation to predict the future of patients from the electronic health records. *Sci Rep* 6:26094
- Mitra B, Craswell N et al (2018) An introduction to neural information retrieval. *Found Trends® Inf Retrieval* 13:1–126
- Nachimovsky H, Tennenholtz M, Kurland O (2025) A multi-agent perspective on modern information retrieval. [arXiv:2502.14796](https://arxiv.org/abs/2502.14796)
- Nogueira R, Cho K (2019) Passage re-ranking with bert. [arXiv:1901.04085](https://arxiv.org/abs/1901.04085)
- Nogueira R, Yang W, Lin J, Cho K (2019) Document expansion by query prediction. [arXiv:1904.08375](https://arxiv.org/abs/1904.08375)
- Pal A, Umapathi LK, Sankarasubbu M (2023) Med-halt: medical domain hallucination test for large language models. [arXiv:2307.15343](https://arxiv.org/abs/2307.15343)
- Qu Y, Ding Y, Liu J, Liu K, Ren R, Zhao WX, Dong D, Wu H, Wang H (2020) Rocketqa: an optimized training approach to dense passage retrieval for open-domain question answering. [arXiv:2010.08191](https://arxiv.org/abs/2010.08191)
- Reimers N, Gurevych I (2019) Sentence-bert: sentence embeddings using siamese bert-networks. [arXiv:1908.10084](https://arxiv.org/abs/1908.10084)
- Roberts K, Alam T, Bedrick S, Demner-Fushman D, Lo K, Soboroff I, Voorhees E, Wang LL, Hersh WR (2020) Trec-covid: rationale

- and structure of an information retrieval shared task for covid-19. *J Am Med Inform Assoc* 27:1431–1436
- Robertson S, Zaragoza H et al (2009) The probabilistic relevance framework: Bm25 and beyond. *Found Trends® Inf Retrieval* 3:333–389
- Santhanam K, Khattab O, Saad-Falcon J, Potts C, Zaharia M (2021) Colbertv2: effective and efficient retrieval via lightweight late interaction. [arXiv:2112.01488](https://arxiv.org/abs/2112.01488)
- Schick T, Dwivedi-Yu J, Dessì R, Raileanu R, Lomeli M, Hambro E, Zettlemoyer L, Cancedda N, Scialom T (2023) Toolformer: language models can teach themselves to use tools. *Adv Neural Inf Process Syst* 36:68539–68551
- Shen Y, Song K, Tan X, Li D, Lu W, Zhuang Y (2023) Hugginggpt: solving ai tasks with chatgpt and its friends in hugging face. *Adv Neural Inf Process Syst* 36:38154–38180
- Singhal K, Azizi S, Tu T, Mahdavi SS, Wei J, Chung HW, Scales N, Tanwani A, Cole-Lewis H, Pfohl S et al (2023) Large language models encode clinical knowledge. *Nature* 620:172–180
- Thakur N, Reimers N, Rücklé A, Srivastava A, Gurevych I (2021) BEIR: a heterogeneous benchmark for zero-shot evaluation of information retrieval models
- Touvron H, Martin L, Stone K, Albert P, Almahairi A, Babaei Y, Bashlykov N, Batra S, Bhargava P, Bhosale S et al (2023) Llama 2: open foundation and fine-tuned chat models. [arXiv:2307.09288](https://arxiv.org/abs/2307.09288)
- Tsatsaronis G, Balikas G, Malakasiotis P, Partalas I, Zschunke M, Alvers MR, Weissenborn D, Krithara A, Petridis S, Polychronopoulos D et al (2015) An overview of the bioasq large-scale biomedical semantic indexing and question answering competition. *BMC Bioinform* 16:138
- Wang C, Ong J, Wang C, Ong H, Cheng R, Ong D (2024) Potential for gpt technology to optimize future clinical decision-making using retrieval-augmented generation. *Ann Biomed Eng* 52:1115–1118
- Wei J, Wang X, Schuurmans D, Bosma M, Xia F, Chi E, Le QV, Zhou D et al (2022) Chain-of-thought prompting elicits reasoning in large language models. *Adv Neural Inf Process Syst* 35:24824–24837
- Xiong G, Jin Q, Lu Z, Zhang A (2024) Benchmarking retrieval-augmented generation for medicine. In: *Findings of the association for computational linguistics ACL 2024*, pp 6233–6251
- Xiong L, Xiong C, Li Y, Tang K-F, Liu J, Bennett P, Ahmed J, Overwijk A (2020) Approximate nearest neighbor negative contrastive learning for dense text retrieval. [arXiv:2007.00808](https://arxiv.org/abs/2007.00808)
- Yao S, Zhao J, Yu D, Du N, Shafran I, Narasimhan K, Cao Y (2023) React: synergizing reasoning and acting in language models. In: *International Conference on Learning Representations (ICLR)*
- Zakka C, Shad R, Chaurasia A, Dalal AR, Kim JL, Moor M, Fong R, Phillips C, Alexander K, Ashley E et al (2024) Almanac—retrieval-augmented language models for clinical medicine. *Nejm ai* 1:A10a2300068
- Zhao P, Zhang H, Yu Q, Wang Z, Geng Y, Fu F, Yang L, Zhang W, Jiang J, Cui B (2024) Retrieval-augmented generation for ai-generated content: a survey. [arXiv:2402.19473](https://arxiv.org/abs/2402.19473)
- Zhao Z, Jin Q, Chen F, Peng T, Yu S (2023) A large-scale dataset of patient summaries for retrieval-based clinical decision support systems. *Sci Data* 10:909

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